ORGANIC RELIEF + RETOPO — JUNGLE RUINS

Canopy doctrine in, sculpt-ready quads out: grid 512, loose edges, hole-fill, quad remesh, textured OBJ+GLB.

FILE **atlas_organic_relief_jungleruins_workflow.json** VERIFIED **2026-07-16**

THE STORY

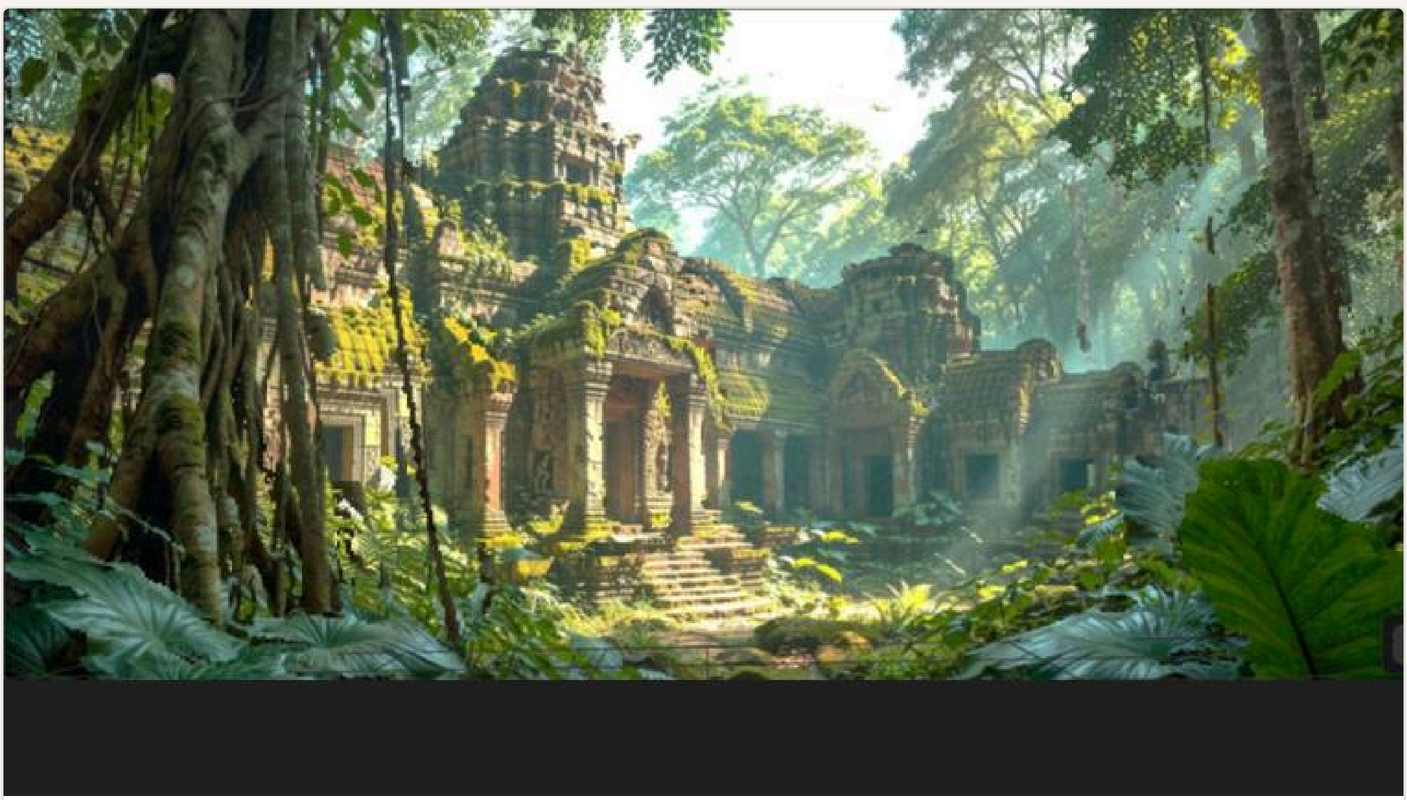
Dense canopy depth is genuinely noisy at small scale. The relief doctrine for organics: raise the grid (512 here) so each quad spans less real-world area, and LOOSEN depth_edge_rel to 1.0 — the default 0.5 reads canopy noise as depth discontinuities and shreds the mesh into holes. An AtlasBoundedBand clips the foreground's own extrusion so monocular 'banana' error can't run the ruins off into the distance.

The export is where this workflow earns its place: export-only hole-fill caps small interior tears (the live projection mesh keeps its deliberate silhouette tears — that doctrine is untouched), then QUAD retopology remeshes down to ~4,000 target vertices with projective UVs regenerated by inverting the camera bake — so the retopologized mesh still imports TEXTURED into Maya, ZBrush, or Blender. build_scene.py reproduces the whole setup in Blender's Script editor.

IN THE VIEWPORT — PROJECTED VS. THE GREY MESH

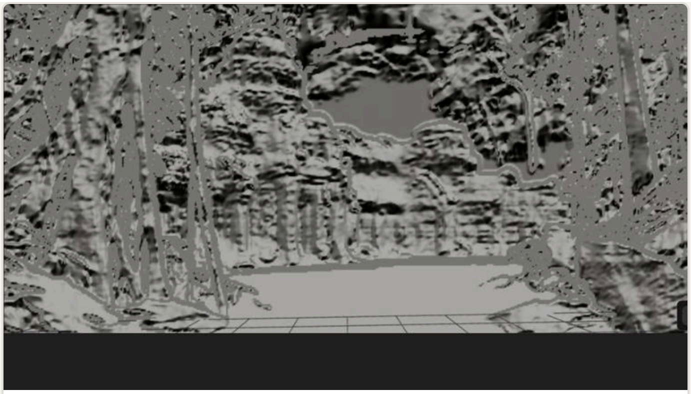
SOURCE PLATE

jungleruins_32bit_acescg.exr · ACEScg float — the single still photograph everything below is reconstructed from.



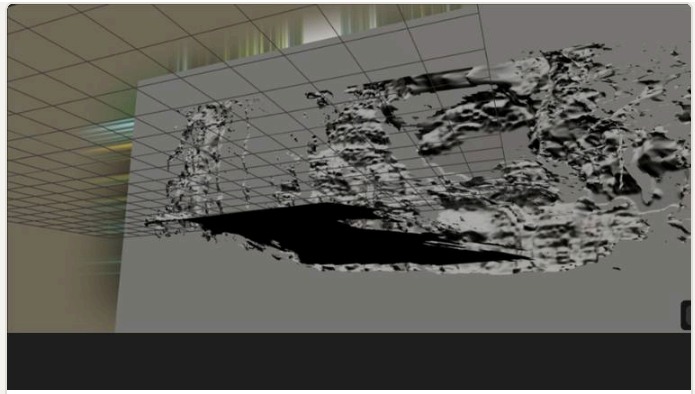
PROJECT ON

The photo reassembled on the recovered camera — texels assigned by ray, perfect from this view.



PROJECT OFF

The same geometry as a grey mesh, recovered view — what the depth-derived surfaces actually are.

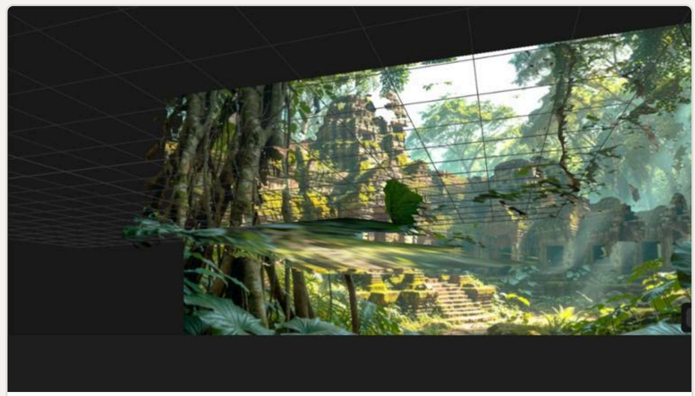


GREY · ORBIT LEFT

Orbited off-axis: the reconstructed cone. Coverage ends where the camera never saw.

GREY · ORBIT RIGHT

The opposite angle — silhouette tears read as deliberate holes, not errors.



BAND BOX

📏 overlay: the translucent box's back face is the metric cutoff plane the layer is clipped at.

NODES ON STAGE

`OCIORead` `AtlasDepthMap` `AtlasDeriveReliefMesh` `AtlasBoundedBand` `AtlasExportReliefMesh` `AtlasExportBlender`

RETUNE ON YOUR PLATE

- **relief_grid / depth_edge_rel — the organic dial pair.** Denser canopy wants a finer grid AND a looser edge threshold. Watch the hole fraction, not the vert count.
- **retopo_method + retopo_target_vertex_count.** quad (pyinstantmeshes) for sculpting; decimate for lightweight look-dev; smooth to relax without changing topology.
- **fill_interior_holes + the depth window.** fill_depth_near/far_m scopes which tears get capped — transcribe AtlasBoundedBand's cutoff_m to fill only inside the subject.

- **Preview before the DCC round-trip.** Wire the export's preview_solve into a viewport — it carries the mesh ACTUALLY written, fills and all.

RUN-VERIFIED

RUN ✓ Quad retopo landed at 4,493 verts (target 4,000); OBJ and GLB import textured — projective UVs regenerated and pinned to 1e-5; build_scene.py ready for Blender.

REQUIREMENTS

`OCIO/EXR env · pyinstantmeshes + fast-simplification (retopo) · [neural]`